

Student API — A to Z Exam Guidance Handbook

Spring Boot · Postman · Git · Docker · Jenkins

Print this before the exam. Follow **Part A** in order (Start → End).

If something breaks, go to **Part C — Problems and Fixing** at the end.

Item	Value
Local folder	C:\Users\rasar\OneDrive\Desktop\SOC
GitHub	https://github.com/Razaara/student-api
Docker Hub	razzara/student-api
Port	8500
Base URL	http://localhost:8500/api/students
DB name	students
DB user / pass	root / root
Java	21
Spring Boot	3.4.5

How this book is organized

Part	What it is
Part A	Project steps Start → End (do in this exact order)
Part B	Quick reference (annotations)
Part C	Problems and Fixing (Git, ports, Spring, Docker, Jenkins)
Part D	One-page final checklist

PART A — Project Steps (Start → End)

Do these steps in order. Do not skip ahead unless the examiner asks.

```
STEP 01 Create Spring Boot project
STEP 02 Project structure + .gitignore
STEP 03 pom.xml dependencies
STEP 04 application.properties
STEP 05 Main class
STEP 06 Entity → Repository → Service → Controller
STEP 07 Start MySQL + run the app
STEP 08 Test with Postman (all 5 endpoints)
STEP 09 Git commit + push to GitHub
```

```
STEP 10 Docker (Dockerfile / compose)
STEP 11 Push image to Docker Hub
STEP 12 Jenkins Pipeline → Build Now → re-test
```

If time is short: Steps 01–08 first → 09 Git → 10 Docker → 12 Jenkins last.

STEP 01 — Create Spring Boot project

1. Open <https://start.spring.io> or IntelliJ → New → Spring Initializr
2. Project: **Maven** · Language: **Java** · Java: **21**
3. Group: `com.example` · Artifact: `studentapi` · Package: `com.example.studentapi`
4. Add dependencies:
 - Spring Web
 - Spring Data JPA
 - MySQL Driver
 - Lombok
 - Validation
5. Generate and extract into `C:\Users\rasar\OneDrive\Desktop\SOC`

If Initializr only offers Boot 4.x, use Boot **3.4.x** if available, or keep parent version `3.4.5` as in this handbook.

Already have the working repo?

```
cd C:\Users\rasar\OneDrive\Desktop
git clone https://github.com/Razaara/student-api.git SOC
cd SOC
```

STEP 02 — Project structure + .gitignore

```
SOC/
├─ pom.xml
├─ Dockerfile
├─ docker-compose.yml
├─ Jenkinsfile
├─ .gitignore
└─ src/main/
    ├─ java/com/example/studentapi/
    │   ├─ StudentApiApplication.java
    │   ├─ entity/Student.java
    │   ├─ repository/StudentRepository.java
    │   ├─ service/StudentService.java
    │   ├─ service/StudentServiceImpl.java
    │   └─ controller/StudentController.java
    └─ resources/application.properties
```

.gitignore

```
target/
.idea/
```

```
*.iml
*.class
.env
*.log
```

STEP 03 — pom.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://maven.apache.org/POM/4.0.0"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
https://maven.apache.org/xsd/maven-4.0.0.xsd">
    <modelVersion>4.0.0</modelVersion>

    <parent>
        <groupId>org.springframework.boot</groupId>
        <artifactId>spring-boot-starter-parent</artifactId>
        <version>3.4.5</version>
        <relativePath/>
    </parent>

    <groupId>com.example</groupId>
    <artifactId>studentapi</artifactId>
    <version>0.0.1-SNAPSHOT</version>
    <name>studentapi</name>
    <description>Student CRUD REST API for SOC practical exam</description>

    <properties>
        <java.version>21</java.version>
    </properties>

    <dependencies>
        <dependency>
            <groupId>org.springframework.boot</groupId>
            <artifactId>spring-boot-starter-web</artifactId>
        </dependency>
        <dependency>
            <groupId>org.springframework.boot</groupId>
            <artifactId>spring-boot-starter-data-jpa</artifactId>
        </dependency>
        <dependency>
            <groupId>com.mysql</groupId>
            <artifactId>mysql-connector-j</artifactId>
            <scope>runtime</scope>
        </dependency>
        <dependency>
            <groupId>org.projectlombok</groupId>
            <artifactId>lombok</artifactId>
            <optional>true</optional>
        </dependency>
```

```

    <dependency>
      <groupId>org.springframework.boot</groupId>
      <artifactId>spring-boot-starter-validation</artifactId>
    </dependency>
    <dependency>
      <groupId>org.springframework.boot</groupId>
      <artifactId>spring-boot-starter-test</artifactId>
      <scope>test</scope>
    </dependency>
  </dependencies>

  <build>
    <plugins>
      <plugin>
        <groupId>org.springframework.boot</groupId>
        <artifactId>spring-boot-maven-plugin</artifactId>
        <configuration>
          <excludes>
            <exclude>
              <groupId>org.projectlombok</groupId>
              <artifactId>lombok</artifactId>
            </exclude>
          </excludes>
        </configuration>
      </plugin>
    </plugins>
  </build>
</project>

```

STEP 04 — application.properties

File: src/main/resources/application.properties

```

spring.application.name=studentapi
spring.datasource.url=jdbc:mysql://localhost:3306/students?createDatabaseIfNotExist=true
spring.datasource.username=root
spring.datasource.password=root
spring.jpa.show-sql=true
spring.jpa.hibernate.ddl-auto=update
server.port=8500

```

Setting	Meaning
createDatabaseIfNotExist=true	Creates DB if missing
ddl-auto=update	Creates/updates tables from entity
server.port=8500	API port (not 8080)
password root	Matches Docker MySQL

STEP 05 — Main class

File: StudentApiApplication.java

```
package com.example.studentapi;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;

@SpringBootApplication
public class StudentApiApplication {

    public static void main(String[] args) {
        SpringApplication.run(StudentApiApplication.class, args);
    }
}
```

All other classes must stay under package `com.example.studentapi...` for component scanning.

STEP 06 — Entity → Repository → Service → Controller

Create in this order.

6.1 Entity — Student.java

```
package com.example.studentapi.entity;

import jakarta.persistence.Entity;
import jakarta.persistence.Id;
import jakarta.persistence.Table;
import lombok.AllArgsConstructor;
import lombok.Data;
import lombok.NoArgsConstructor;

@Data
@Entity
@Table(name = "student")
@AllArgsConstructor
@NoArgsConstructor
public class Student {

    @Id
    private String id;
    private String name;
    private String email;
    private Integer age;
}
```

Use `jakarta.persistence` (Boot 3). Send `id` in POST body (e.g. `AS2023000`).

6.2 Repository — StudentRepository.java

```
package com.example.studentapi.repository;

import com.example.studentapi.entity.Student;
import org.springframework.data.jpa.repository.JpaRepository;

public interface StudentRepository extends JpaRepository<Student, String> {
}
```

6.3 Service interface — StudentService.java

```
package com.example.studentapi.service;

import com.example.studentapi.entity.Student;
import java.util.List;

public interface StudentService {
    Student save(Student student);
    List<Student> getAll();
    Student getById(String id);
    Student update(String id, Student student);
    void delete(String id);
}
```

6.4 Service impl — StudentServiceImpl.java

```
package com.example.studentapi.service;

import com.example.studentapi.entity.Student;
import com.example.studentapi.repository.StudentRepository;
import lombok.RequiredArgsConstructor;
import org.springframework.stereotype.Service;
import java.util.List;

@Service
@RequiredArgsConstructor
public class StudentServiceImpl implements StudentService {

    private final StudentRepository studentRepository;

    @Override
    public Student save(Student student) {
        return studentRepository.save(student);
    }

    @Override
    public List<Student> getAll() {
        return studentRepository.findAll();
    }
}
```

```

    }

    @Override
    public Student getById(String id) {
        return studentRepository.findById(id)
            .orElseThrow(() -> new RuntimeException("Student not found with id: " +
id));
    }

    @Override
    public Student update(String id, Student student) {
        Student existing = getById(id);
        existing.setName(student.getName());
        existing.setEmail(student.getEmail());
        existing.setAge(student.getAge());
        return studentRepository.save(existing);
    }

    @Override
    public void delete(String id) {
        studentRepository.deleteById(id);
    }
}

```

6.5 Controller — StudentController.java

```

package com.example.studentapi.controller;

import com.example.studentapi.entity.Student;
import com.example.studentapi.service.StudentService;
import lombok.RequiredArgsConstructor;
import org.springframework.http.HttpStatus;
import org.springframework.http.ResponseEntity;
import org.springframework.web.bind.annotation.*;
import java.util.List;

@RestController
@RequestMapping("/api/students")
@RequiredArgsConstructor
public class StudentController {

    private final StudentService studentService;

    @PostMapping
    public ResponseEntity<Student> create(@RequestBody Student student) {
        Student saved = studentService.save(student);
        return new ResponseEntity<>(saved, HttpStatus.CREATED);
    }

    @GetMapping
    public ResponseEntity<List<Student>> getAll() {

```

```

        return ResponseEntity.ok(studentService.getAll());
    }

    @GetMapping("/{id}")
    public ResponseEntity<Student> getById(@PathVariable String id) {
        return ResponseEntity.ok(studentService.getById(id));
    }

    @PutMapping("/{id}")
    public ResponseEntity<Student> update(@PathVariable String id, @RequestBody Student
student) {
        return ResponseEntity.ok(studentService.update(id, student));
    }

    @DeleteMapping("/{id}")
    public ResponseEntity<Void> delete(@PathVariable String id) {
        studentService.delete(id);
        return ResponseEntity.noContent().build();
    }
}

```

Endpoints (remember these)

Method	Path	Success
POST	/api/students	201
GET	/api/students	200
GET	/api/students/{id}	200
PUT	/api/students/{id}	200
DELETE	/api/students/{id}	204

STEP 07 — Start MySQL + run the app

7.1 Start MySQL (Docker)

```

cd C:\Users\rasar\OneDrive\Desktop\SOC
docker compose up -d mysql
docker ps

```

Wait until `student-mysql` is **healthy**.

7.2 Run Spring Boot

```

mvn clean package -DskipTests
mvn spring-boot:run

```

Console must show:

```
Tomcat started on port(s): 8500
Started StudentApiApplication
```

Port busy? Jump to **Part C** → **Ports**.

STEP 08 — Test with Postman

Base URL: `http://localhost:8500/api/students`

Body: **raw** → **JSON**

8.1 POST — Create (201)

POST `http://localhost:8500/api/students`

```
{
  "id": "AS2023000",
  "name": "John Doe",
  "email": "john@example.com",
  "age": 21
}
```

8.2 GET — All (200)

GET `http://localhost:8500/api/students`

8.3 GET — By id (200)

GET `http://localhost:8500/api/students/AS2023000`

8.4 PUT — Update (200)

PUT `http://localhost:8500/api/students/AS2023000`

```
{
  "name": "John Updated",
  "email": "john.updated@example.com",
  "age": 22
}
```

8.5 DELETE (204)

DELETE `http://localhost:8500/api/students/AS2023000`

Test order

POST (201) → GET all (200) → GET by id (200) → PUT (200) → DELETE (204)

STEP 09 — Git commit + push to GitHub

```
cd C:\Users\rasar\OneDrive\Desktop\SOC
git init
git add .
git commit -m "Initial commit: Student CRUD API"
git branch -M main
git remote add origin https://github.com/Razaara/student-api.git
git push -u origin main
```

Everyday:

```
git status
git add .
git commit -m "Add update endpoint"
git push
```

Shared lab PC (before push)

1. Credential Manager → Windows Credentials → remove `github.com`
2. Then:

```
git config user.name "Your Name"
git config user.email "your-email@example.com"
git remote -v
git push -u origin main
```

Login with **your** GitHub account when prompted.

This project repo: <https://github.com/Razaara/student-api>

*Git errors? Jump to **Part C** → **Git**.*

STEP 10 — Docker

10.1 Dockerfile

```
FROM maven:3.9-eclipse-temurin-21 AS build
WORKDIR /app
COPY pom.xml .
COPY src ./src
RUN mvn -B clean package -DskipTests

FROM eclipse-temurin:21-jre-alpine
WORKDIR /app
COPY --from=build /app/target/*.jar app.jar
EXPOSE 8500
ENTRYPOINT ["java", "-jar", "app.jar"]
```

10.2 Build & run one container (MySQL already running)

```
docker build -t student-api .
docker run -d -p 8500:8500 --name student-api-container ^
  -e SPRING_DATASOURCE_URL=jdbc:mysql://host.docker.internal:3306/students?
createDatabaseIfNotExist=true ^
  -e SPRING_DATASOURCE_USERNAME=root ^
  -e SPRING_DATASOURCE_PASSWORD=root ^
  student-api

docker logs -f student-api-container
```

10.3 docker-compose.yml (MySQL + API together)

```
services:
  mysql:
    image: mysql:8.0
    container_name: student-mysql
    restart: always
    environment:
      MYSQL_ROOT_PASSWORD: root
      MYSQL_DATABASE: students
    ports:
      - "3306:3306"
    volumes:
      - mysql_data:/var/lib/mysql
    healthcheck:
      test: ["CMD", "mysqladmin", "ping", "-h", "localhost", "-uroot", "-proot"]
      interval: 10s
      timeout: 5s
      retries: 10
      start_period: 20s

  springboot:
    build: .
    container_name: student-api
    restart: on-failure
    ports:
      - "8500:8500"
    environment:
      SPRING_DATASOURCE_URL: jdbc:mysql://mysql:3306/students?createDatabaseIfNotExist=true
      SPRING_DATASOURCE_USERNAME: root
      SPRING_DATASOURCE_PASSWORD: root
    depends_on:
      mysql:
        condition: service_healthy

volumes:
  mysql_data:
```

```
docker compose up -d --build
docker compose ps
docker compose logs -f springboot
docker compose down
```

Compose DB host = service name `mysql` . Single `docker run` DB host = `host.docker.internal` .

STEP 11 — Push image to Docker Hub

Docker user: `razzara`

```
docker login
docker tag student-api:latest razzara/student-api:latest
docker push razzara/student-api:latest
```

```
docker pull razzara/student-api:latest
```

STEP 12 — Jenkins Pipeline

12.1 One-time tools

Manage Jenkins → Tools

- Maven name must be exactly: `Maven`
- Java 21 on the machine
- Docker Desktop running

12.2 Create job

1. <http://localhost:8080> → Login
2. **New Item** → `student-api` → **Pipeline**
3. Pipeline script from SCM → Git
4. URL: `https://github.com/Razaara/student-api.git`
5. Branch: `*/main` · Script Path: `Jenkinsfile`
6. Save → **Build Now**

12.3 Before Build Now

```
docker compose up -d mysql
docker stop student-api student-api-container 2>NUL
docker rm student-api student-api-container 2>NUL
```

12.4 Jenkinsfile (working)

```
pipeline {
    agent any

    tools {
```

```

    maven 'Maven'
}

environment {
    IMAGE_NAME = 'student-api'
    CONTAINER_NAME = 'student-api-container'
    DB_URL = 'jdbc:mysql://host.docker.internal:3306/students?
createDatabaseIfNotExist=true'
    DB_USER = 'root'
    DB_PASS = 'root'
}

stages {
    stage('Checkout') {
        steps {
            git branch: 'main', url: 'https://github.com/Razaara/student-api.git'
        }
    }

    stage('Build with Maven') {
        steps {
            bat 'mvn clean package -DskipTests'
        }
    }

    stage('Build Docker Image') {
        steps {
            bat 'docker build -t %IMAGE_NAME% .'
        }
    }

    stage('Stop Old Container') {
        steps {
            bat 'docker stop %CONTAINER_NAME% 2>NUL || exit /b 0'
        }
    }

    stage('Remove Old Container') {
        steps {
            bat 'docker rm %CONTAINER_NAME% 2>NUL || exit /b 0'
        }
    }

    stage('Run New Container') {
        steps {
            bat '''
                docker run -d ^
                -p 8500:8500 ^
                --name %CONTAINER_NAME% ^
                -e SPRING_DATASOURCE_URL=%DB_URL% ^
                -e SPRING_DATASOURCE_USERNAME=%DB_USER% ^
                -e SPRING_DATASOURCE_PASSWORD=%DB_PASS% ^
            '''
        }
    }
}

```

```

        %IMAGE_NAME%
    ...
}
}

stage('Verify') {
    steps {
        bat '''
            setlocal EnableDelayedExpansion
            set OK=0
            for /L %i in (1,1,18) do (
                curl.exe -f -s -o NUL http://localhost:8500/api/students
                if !ERRORLEVEL! EQU 0 (
                    echo API is UP
                    set OK=1
                    goto :done
                )
                echo Waiting for API... attempt %i
                ping -n 6 127.0.0.1 >NUL
            )
            :done
            if !OK! NEQ 1 (
                echo API did not become ready
                docker logs %CONTAINER_NAME%
                exit /b 1
            )
            curl.exe -s http://localhost:8500/api/students
            echo.
        ...
    }
}

post {
    success {
        echo 'Deployment successful. API: http://localhost:8500/api/students'
    }
    failure {
        echo 'Pipeline failed – check console output.'
        bat 'docker logs %CONTAINER_NAME% 2>NUL || exit /b 0'
    }
}
}

```

Pipeline stages (examiner view)

Checkout → Maven → Docker Build → Stop → Remove → Run → Verify

12.5 After SUCCESS — re-test in Postman

Open `http://localhost:8500/api/students` and run POST/GET again.

PART B — Quick Reference (Annotations)

Annotation	Use
@SpringBootApplication	Main class
@RestController	REST controller
@RequestMapping	Base path /api/students
@PostMapping / @GetMapping / @PutMapping / @DeleteMapping	HTTP methods
@PathVariable	{id} from URL
@RequestBody	JSON → Java object
@Service	Service bean
@Entity / @Id / @Table	JPA entity
@Data	Lombok getters/setters
@RequiredArgsConstructor	Constructor injection

PART C — Problems and Fixing (FINAL)

Use this part only when something fails. Find your error → apply the fix → return to Part A.

C1. Quick problem index

Area	Go to
Spring Boot / MySQL / Postman	C2
Git / GitHub	C3
Port already in use (8500 / 3306 / 8080)	C4
Docker / Docker Hub	C5
Jenkins	C6

C2. Spring Boot / MySQL / Postman problems

Problem	Fix
Communications link failure	<code>docker compose up -d mysql — wait until healthy</code>
Access denied for user 'root'	Set password to root in <code>application.properties</code>
Unknown database	Keep <code>createDatabaseIfNotExist=true</code>

Table doesn't exist	Keep <code>spring.jpa.hibernate.ddl-auto=update</code>
Whitelabel 404	Check URL = <code>/api/students</code> and controller mapping
400 Bad Request on POST	Valid JSON + include "id" field; use Postman raw JSON
500 on GET missing id	Expected with sample code (<code>RuntimeException</code>)
<code>javax.persistence</code> errors	Change imports to <code>jakarta.persistence</code>
Lombok getters not found	Install Lombok plugin + enable annotation processing
App starts but Postman ECONNREFUSED	Confirm console: Tomcat started on port(s): 8500

C3. Common Git issues

Problem	Cause	Fix
fatal: not a git repository	Wrong folder / no init	<code>cd project → git init</code>
remote origin already exists	Origin already added	<code>git remote set-url origin https://github.com/Razaara/student-api.git</code>
Updates rejected (non-fast-forward)	Remote ahead of local	<code>git pull origin main</code> then <code>git push</code>
Authentication failed	GitHub blocks password	Use PAT / Git Credential Manager login
Wrong GitHub account (lab PC)	Old Windows credentials	Credential Manager → remove <code>github.com</code> → push → login again
Permission denied (publickey)	SSH not set up	Use HTTPS remote URL
Everything up-to-date but GitHub empty	Forgot commit	<code>git add . → git commit -m "msg" → git push</code>
Committed target/	Missing <code>.gitignore</code>	<code>Add target/ → git rm -r --cached target/ → commit → push</code>
Merge conflict <<<<<<	Same lines edited twice	Edit file → remove markers → <code>git add . → git commit</code>
Unrelated histories	New local + existing remote	<code>git pull origin main --allow-unrelated-histories</code>
Please tell me who you are	Identity missing	<code>git config user.name "..."</code> and <code>user.email "..."</code>
Detached HEAD	Checked out a commit	<code>git checkout main</code>

Git recovery commands

```
git status
git remote -v
git log --oneline -5

git remote set-url origin https://github.com/Razaara/student-api.git
git pull origin main
git push origin main

git rm -r --cached target/
git add .
git commit -m "Stop tracking target folder"
git push
```

Shared PC Git checklist

1. Credential Manager → remove github.com
2. git config user.name "Your Name"
3. git config user.email "your-email@example.com"
4. git remote -v (must be YOUR repo)
5. git add . → commit → push
6. Login with YOUR GitHub account

`user.name` / `user.email` do **not** log you into GitHub. Login/PAT does.

C4. Ports already in use — how to fix

Port	Used by
8500	Spring Boot API
3306	MySQL
8080	Jenkins

Symptoms

- Port 8500 was already in use
- Bind for 0.0.0.0:8500 failed: port is already allocated

Find what uses the port

```
netstat -ano | findstr :8500
```

Last column = **PID**.

Fix A — Stop Docker containers (most common)

```
docker ps
docker stop student-api student-api-container student-mysql 2>NUL
```

```
docker rm student-api student-api-container student-mysql 2>NUL
cd C:\Users\rasar\OneDrive\Desktop\SOC
docker compose down
```

Then restart what you need:

```
docker compose up -d mysql
```

Fix B — Kill Java / local process

```
netstat -ano | findstr :8500
taskkill /PID <PID> /F
```

Or:

```
Get-NetTCPConnection -LocalPort 8500 -ErrorAction SilentlyContinue |
    ForEach-Object { Stop-Process -Id $_.OwningProcess -Force -ErrorAction SilentlyContinue }
```

Fast fix — port 8500

```
docker stop student-api student-api-container 2>NUL
docker rm student-api student-api-container 2>NUL
netstat -ano | findstr :8500
taskkill /PID <PID> /F
mvn spring-boot:run
```

Fast fix — port 3306

```
docker stop student-mysql 2>NUL
docker rm student-mysql 2>NUL
docker compose up -d mysql
```

Fast fix — port 8080 (Jenkins)

```
netstat -ano | findstr :8080
Get-Service Jenkins
Start-Service Jenkins
```

Last resort — change app port

```
server.port=8501
```

Then Postman: `http://localhost:8501/api/students`

Prefer freeing the port so handbook URLs stay `8500` .

C5. Docker / Docker Hub problems

Problem	Fix
Cannot connect to Docker daemon	Start Docker Desktop
Port already allocated	See C4
Communications link failure in container	Use <code>host.docker.internal</code> (docker run) or <code>mysql</code> (compose)
Container exits immediately	<code>docker logs <name></code> — read startup error
Push denied	<code>docker login</code> then tag <code>razzara/student-api</code>
Old code still running	<code>docker build --no-cache -t student-api .</code>
No space left	<code>docker system prune -a</code>

C6. Jenkins problems

Problem	Fix
<code>mvn: command not found</code>	Tools name must be exactly Maven
<code>docker: command not found</code>	Install/start Docker; restart Jenkins
Checkout auth failed	Public repo URL correct; or add GitHub credentials
Port 8500 conflict on Run stage	Stop old containers (see C4) before Build Now
Container can't reach MySQL	Keep <code>host.docker.internal</code> in Jenkinsfile env
Pipeline red / Verify failed	Open Console Output + <code>docker logs student-api-container</code>

C7. HTTP status meanings

Code	Meaning
200	OK (GET / PUT)
201	Created (POST)
204	No Content (DELETE)
400	Bad request / bad JSON
404	Wrong URL / not mapped
500	Server exception

PART D — One-Page Final Checklist

Start → End command strip

```
# STEP 07
docker compose up -d mysql
mvn spring-boot:run

# STEP 08 — Postman: POST → GET → GET id → PUT → DELETE

# STEP 09
git add .
git commit -m "Student CRUD API"
git push -u origin main

# STEP 10-11
docker build -t student-api .
docker compose up -d --build
docker tag student-api razzara/student-api
docker push razzara/student-api

# STEP 12
# http://localhost:8080 → student-api → Build Now
```

Remember

- Follow **Part A steps in order**
- Port **8500** everywhere
- Include **id** in POST body
- `jakarta.persistence` not `javax`
- Compose DB host = `mysql`
- Jenkins DB host = `host.docker.internal`
- GitHub: **Razaara** · Docker: **razzara**
- Broken? → **Part C Problems and Fixing**

Viva quick answers

- `@Component` vs `@Service` vs `@Repository` — same mechanism, different layer meaning
- Dependency Injection — Spring injects beans (constructor / `@RequiredArgsConstructor`)
- `JpaRepository` vs `CrudRepository` — Jpa adds paging/sorting helpers
- Image vs Container — image = template; container = running instance
- Declarative pipeline — `pipeline { stages { ... } }` in Jenkinsfile

End of handbook.

Project: `C:\Users\rasar\OneDrive\Desktop\SOC`

Repo: <https://github.com/Razaara/student-api>